FujiNet for the Magnavox Odyssey 2 / Philips Videopac, Rev0: an RP2354B cartridge on the 8048's bus + ESP32-S3.

The cart serves the console's fetches in software (fujinet-firmware pico/o2): /PSEN low reads the cart, the RP drives DB0-DB7 while it lasts and lets go otherwise; the console writes the mailbox with /WR while P14 is low; P10 / P11 select the bank.

cart-bus	the 30-finger edge, 330R into the RP2354B's GPIO0-28 (PicoPAC map), pin B's pull-down, the console-5V sense divider, the activity LED, the self-test and debug-UART pads
rp-core	the RP2354B's supplies and decoupling, core regulator, crystal, SWD, RUN / BOOTSEL / RESET, its USB out to the S3
fujinet	the ESP32-S3 (the FujiNet, the RP's USB host), microSD, WS2812 status light
usb	USB-C, the USBLC6 ESD array, the CP2102N bridge and esptool's auto-program pair
power	+5V = the TPS2116 mux (USB VBUS first, console 5 V second), the RP's LDO (+3V3_RP), the S3's buck (+3V3)

Rails (GND, CONS_5V, VBUS, +5V, +3V3, +3V3_RP, DVDD) join by their power symbols.

Firmware contract (tools/check_nets.py checks this netlist against it): fujinet-firmware pico/o2 picopac_cart.c (A0-A11 GPIO0-11, P11 12, P10 13, D0-D7 GPIO14-21, /PSEN 22, /WR 23, P14 24, LED 25, E 26, T0 27); this board adds pin B on GP28, self-test GP29, console 5 V sense GP40, debug UART GP44 / 45. The S3: fujiversal-o2.h (SD 38-41, UART0 43 / 44, WS2812 48, USB host 19 / 20) + RUN / BOOTSEL on IO4 / IO5 and SD card detect on IO42 (README: firmware changes).

power

File: power.kicad_sch

fujinet

File: fujinet.kicad_sch

rp-core

File: rp-core.kicad_sch

usb

File: usb.kicad_sch

cart-bus

File: cart-bus.kicad_sch

the console cartridge slot (the 30-finger edge) is on cart-bus

Rev0 2026-10-30: RP2354B + ESP32-S3; firmware pico/o2 picopac_cart.c pin map
CERN-OHL-W-2.0; RP2350 regulator corner after Raspberry Pi RP2350A minimal design (MIT)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

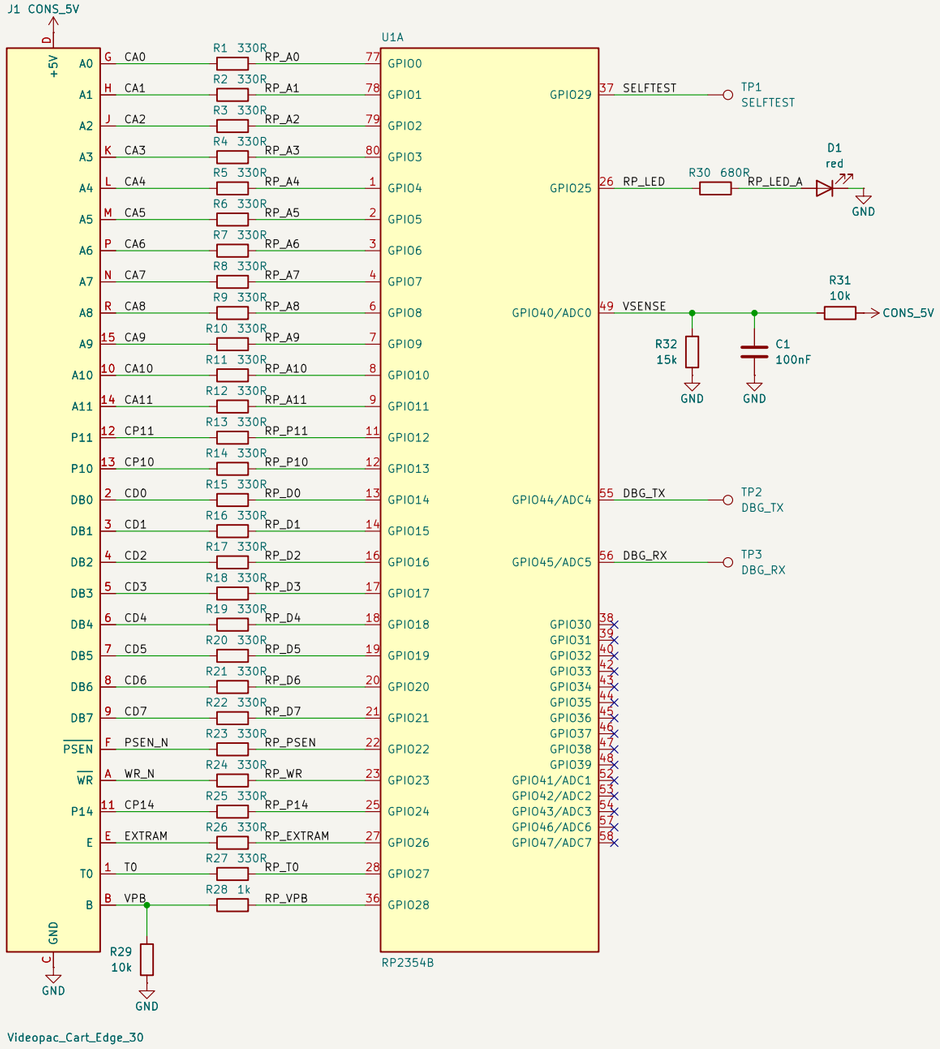
FujiNet

Sheet: /
File: FujiNet-O2-Rev0.kicad_sch

Title: FujiNet Odyssey 2 / Videopac Rev0

Size: A3	Date: 2026-10-30	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 1/6

The Odyssey 2 / Videopac cartridge port (Dan Boris, O2 Tech Specs 2.0): 1–15 on the parts face, A–R behind them.
A0–A11 latched, DB0–DB7 the 8048 bus, /PSEN reads the cart, /WR writes (off while P16 is high), P10 / P11 the
bank lines, P14 the write select, E the external-RAM enable, T0 the 8048 test input (cart -> console).
Every line reaches the RP2354B through 330R, in the PicoPAC map (picopac_cart.c): GPIO0–11 A0–A11, 12 P11,
13 P10, 14–21 DB0–DB7, 22 /PSEN, 23 /WR, 24 P14, 26 E, 27 T0. GPIO0–39 are 5 V tolerant while IOVDD is up.
B: GND on a G7000; on a Videopac+ low shows the 8245 everywhere -- 10k down, 1k from GP28 to raise it.
GP40 senses the console 5 V (10k / 15k = 3.0 V); GP25 drives the LED; GP29 / GP44 / GP45 are bring-up pads.



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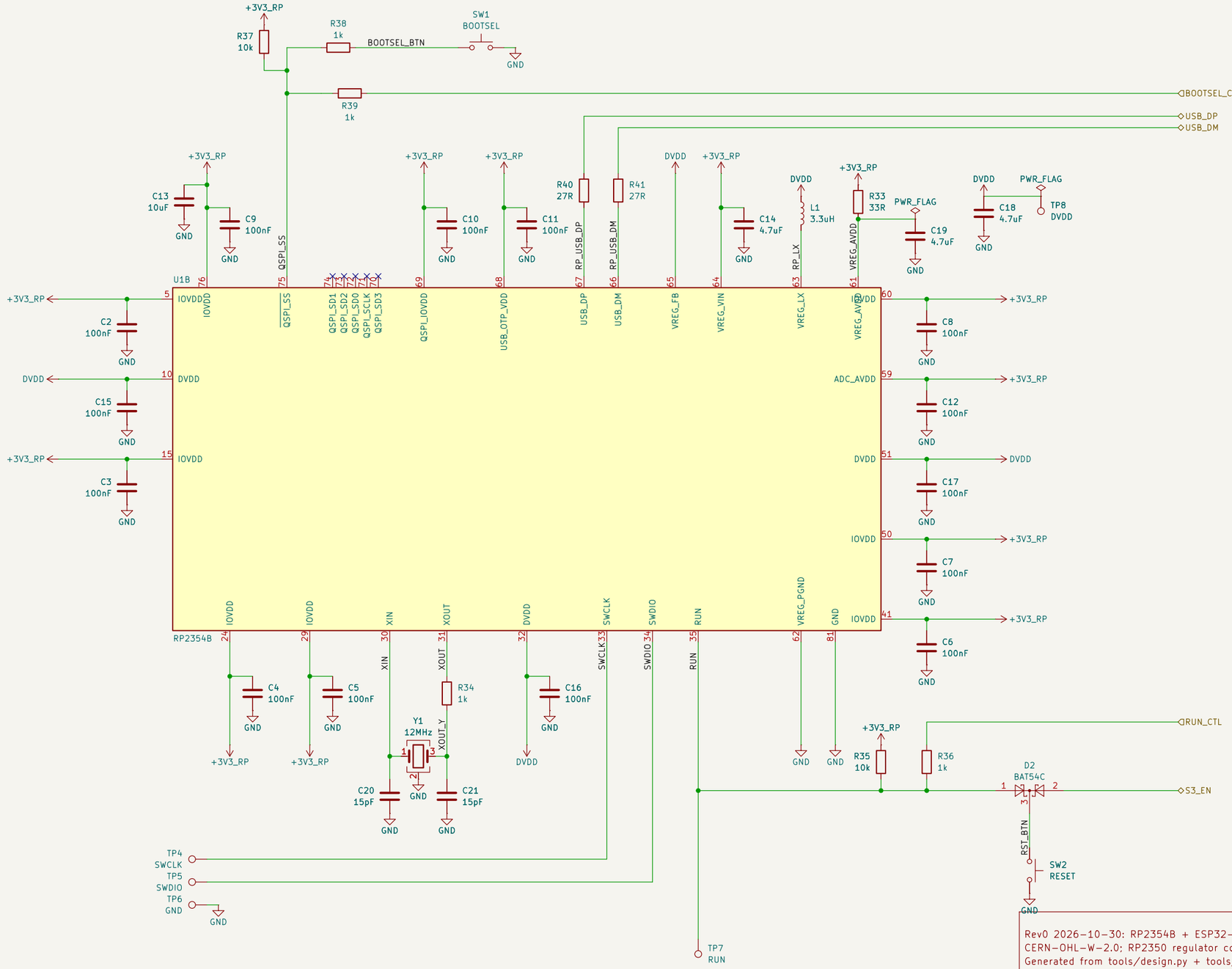
FujiNet

Sheet: /cart-bus/
File: cart-bus.kicad_sch

Title: FujiNet Odyssey 2 / Videopac Rev0 — Cart bus: edge, 330R, RP2354B

Size: A3	Date: 2026-10-30	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 2/6

Supplies (RP2350 datasheet 6.3.7): IOVDD, QSPI_IOVDD, USB_OTP_VDD, ADC_AVDD and VREG_VIN on +3V3_RP, the AP2112K LDO that rises with +5V; the core regulator switches VREG_LX through 3.3 uH onto DVDD (VREG_FB senses it), its analog supply VREG_AVDD through 33R / 4.7 uF. One 100 nF at every supply pin, drawn at the pin it serves. RESET resets the RP2354B and the ESP32-S3 together (D2 steers RST_BTN onto RUN and S3_EN): the way back to the FujiNet's CONFIG. Hold BOOTSEL while pressing RESET for the RP's USB boot ROM; the S3 does both itself: IO4 (RUN_CTL) low resets the RP, IO5 (BOOTSEL_CTL) low through that reset selects BOOTSEL. USB: 27R at the RP, the S3 is the host. SWD and RUN on bring-up pads. QSPI_SD0-3 / SCLK unused: the 2 MB flash is in the package.



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FujiNet

Sheet: /rp-core/
File: rp-core.kicad_sch

Title: FujiNet Odyssey 2 / Videopac Rev0 - RP2354B core, crystal, RUN, SW

Size: A3	Date: 2026-10-30	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 3/6

ESP32-S3-WROOM-1-N16R8: the FujiNet (fujiversal-o2.h pin map), on +3V3 (the buck).
EN: 10k / 1 uF power-on delay. The S3_RST button, the cart RESET
(rp-core, through the BAT54C) and the USB bridge's auto-program pair
(usb sheet) all pull it low; S3_BOOT and the same pair pull IO0.
IO4 / IO5 drive the RP2354B's RUN and BOOTSEL (every on-cart S3's
pins); IO19 / IO20, the S3's own USB, host the RP's.
microSD in SPI mode, 10k pull-ups on CS, DO, DAT1, DAT2 and CD.
WS2812B-2020 on +5V: its data input (VIH 2.7 V) takes the S3's 3.3 V.
The 23 spare GPIOs (right) are not connected.

+3V3
R42
10k
S3_EN
SW3
S3_RST
GND
C24
1uF
GND
+3V3
C23
100nF
GND
C22
22uF
GND
U2
EN
3V3
TXD0
RXD0
S3_TXD
S3_RXD
IO1
IO2
IO3
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO21
IO35
IO36
IO37
IO45
IO46
IO47
S3_IO0
IO0
SW4
S3_BOOT
GND
RUN_CTL
BOOTSEL_CTL
IO4
IO5
USB_DP
USB_DM
USB_D+
USB_D-
ESP32-S3-WROOM-1-N16R8
IO48
GND
IO42
IO41
IO40
IO39
IO38
LED_STRIP
GND
+3V3
RN1C
4x10k
J2
SD_DAT1
SD_DAT2
SD_MOSI
SD_SCK
SD_MISO
SD_CS
SD_CD
DAT1
DAT2
CMD/DI
CLK
DAT0/DO
DAT3/CS
CD
VDD
VSS
microSD
+3V3
R43
10k
+3V3
+3V3
+5V
C26
100nF
GND
DOUT
VDD
DIN
VSS
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R44
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+3V3
C25
10uF
GND
GND

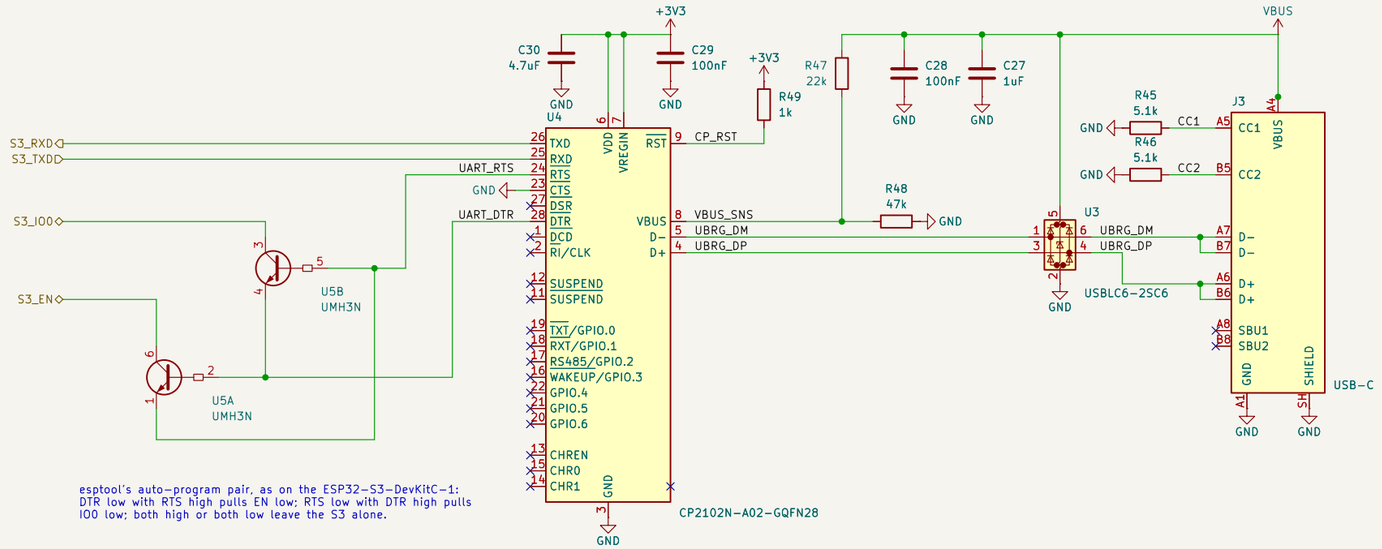
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FujiNet

Sheet: /fujinet/
File: fujinet.kicad_sch

Title: FujiNet Odyssey 2 / Videopac Rev0 — FujiNet (ESP32-S3), microSD

Size: A3	Date: 2026-10-30	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 4/6

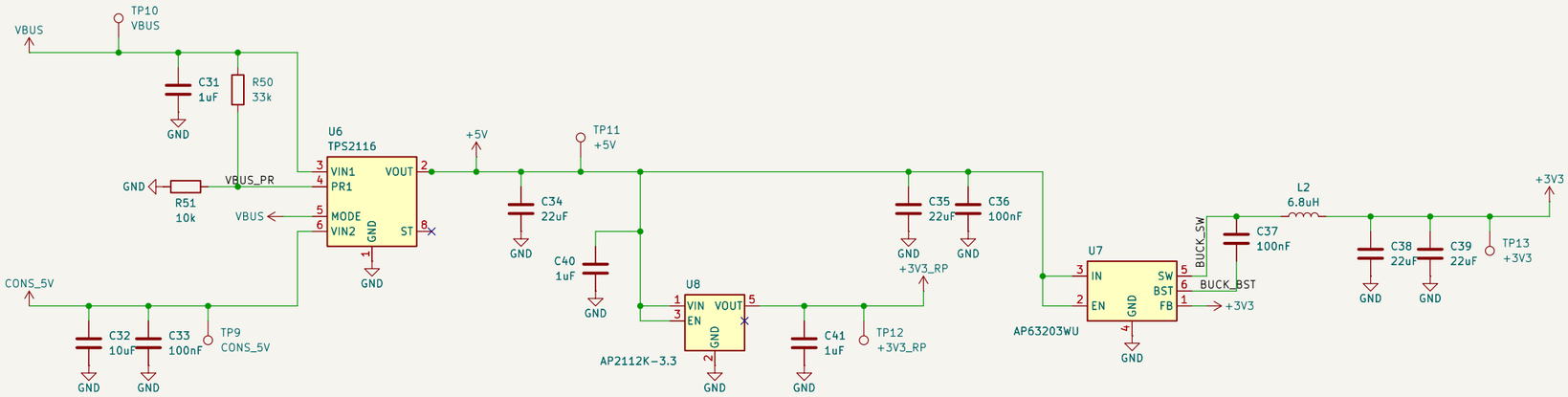


Rev0 2026-10-30: RP2354B + ESP32-S3; firmware pico/o2 picopac_cart.c pin map
CERN-OHL-W-2.0; RP2350 regulator corner after Raspberry Pi RP2350A minimal design (MIT)
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FujiNet

Sheet: /usb/		
File: usb.kicad_sch		
Title: FujiNet Odyssey 2 / Videopac Rev0 — USB-C + CP2102N		
Size: A3	Date: 2026-10-30	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 5/6

Power tree: +5V is the TPS2116 power mux's output (U6), USB first, the console's 5 V second.
  VBUS   USB-C (usb sheet) -> VIN1, the priority input: MODE on VIN1, PR1 from R50 33k / R51 10k, so VBUS
            above VREF x 4.3 (3.96-4.64 V) runs the cart and the console's supply delivers nothing.
  CONS_5V edge pin D, the console's 5 V (spare current not published: prefer USB) -> VIN2, without USB.
  37-60 mOhm either way: reverse current blocking: neither source back-feeds the other, so a cart on USB
  never powers a console that is off. ST is not used.
+5V feeds the WS2812, U8 AP2112K -> +3V3_RP (every RP2354B supply pin: it rises with +5V, so the 5 V-tolerant pads are never unpowered with 5 V on them) and U7 AP63203 -> +3V3 (the ESP32-S3, the CP2102N and the microSD; WiFi bursts ~400 mA; it waits out UVLO, then soft-starts over 4 ms). Without USB the cart adds ~145 mA average, ~370 mA peak to the console's supply (tools/audit/margins.py): prefer USB power. TP10, TP9, TP11, TP12 and TP13 are bring-up pads.



PWR_FLAG: these rails reach their loads only through passive parts (the edge fingers, USB-C, the buck's inductor).



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FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet Odyssey 2 / Videopac Rev0 - Power: mux, S3 buck, RP LDO

Size: A3	Date: 2026-10-30	**Rev: 0**
KiCad E.D.A. 10.0.7		Id: 6/6